

Unit 3 Notes

Combinatorics

Counting, recurrence relations,
and generating functions

These notes begin with elementary counting and build toward systematic methods for solving recurrence relations and extracting coefficients from generating functions. Each section explains the notation, develops the reasoning, and supplies practice with a worked solution.

The coverage follows the supplied syllabus: the pigeonhole principle; homogeneous and nonhomogeneous linear recurrence relations with constant coefficients; applications of recurrence relations; and generating functions. Counting prerequisites are introduced where they support these topics.

There are 52 numbered teaching sections, extra topic-wise practice banks, and cumulative questions with explained answers. All diagrams are black and white. Learn the method, follow the example, and attempt the questions before reading the solutions.

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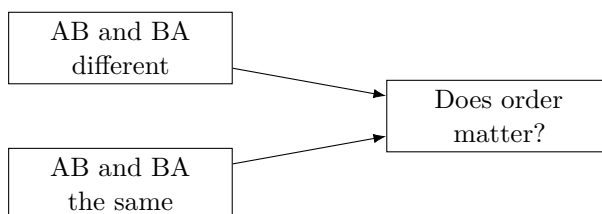
1 Counting as a mathematical model

Combinatorics studies discrete arrangements, selections, and structures. Before applying a formula, identify what counts as one outcome, which outcomes are considered different, and what restrictions are imposed. Most counting mistakes begin with a mismatch between the formula and the objects being counted.

For example, selecting two students for an unordered committee is different from assigning a president and a secretary. In the committee, exchanging the students produces the same selection. In the officer assignment, exchanging them changes the outcome.

Main idea. A counting argument should explain a correspondence. If a procedure produces each permitted outcome exactly once, counting the choices in that procedure counts the objects. If outcomes appear repeatedly or some cannot be produced, the procedure must be corrected.

We use $|S|$ for the number of elements in a finite set S . The notation $\{1, 2, \dots, n\}$ denotes the integers from 1 through n . In sequence discussions, $\mathbb{N}$ will mean the nonnegative integers $0, 1, 2, \dots$; a different starting index will be stated explicitly.



Ask this before choosing a counting formula.

Worked example. How many two-symbol strings can be made from $\{A, B, C\}$ when repetition is allowed? There are three choices for each position, giving nine strings: $AA, AB, AC, BA, BB, BC, CA, CB, CC$. The strings AB and BA are distinct because position matters.

Practice. How many unordered two-element subsets can be selected from $\{A, B, C\}$? Why is the answer not nine?

Solution. The subsets are $\{A, B\}$, $\{A, C\}$, and $\{B, C\}$, so there are three. A subset cannot repeat an element, and its order is irrelevant. These two changes remove both the repeated-letter strings and the distinction between opposite orders.

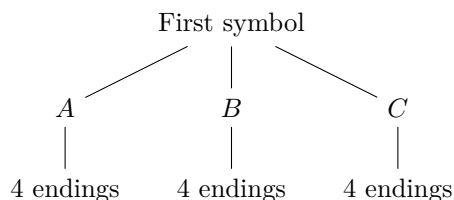
Common error. State whether repetition is permitted and whether order matters before calculating. These are properties of the problem, not optional settings chosen after a formula has been selected.

2 The sum and product principles

The sum principle counts disjoint alternatives. If one class has a outcomes and another disjoint class has b , choosing an outcome from either class gives $a + b$ possibilities. Disjointness matters because an overlapping outcome would otherwise be counted twice.

The product principle counts stages. If there are a choices for the first stage and exactly b choices for the second after each first choice, there are ab complete outcomes. The available second-stage objects may change with the first choice; what matters for this simple product is that their number stays b .

Worked example. A code uses a letter from $\{A, B, C\}$ followed by a digit from $\{1, 2, 3, 4\}$. There are $3 \cdot 4 = 12$ codes. If a different allowed code consists only of one digit, these four additional codes are disjoint from the two-symbol codes, giving $12 + 4 = 16$ in total.



Practice. A student chooses one of three mathematics courses and one of two computing courses. A particular mathematics course conflicts with one particular computing course. How many valid pairs remain?

Solution. Without the restriction there are $3 \cdot 2 = 6$ pairs. Exactly one pair is forbidden, so five remain. We cannot multiply by a uniform number of allowed computing choices after every mathematics choice: one branch has one allowed choice while the other two have two. Adding branch counts, $1 + 2 + 2 = 5$, gives the same result.

Common error. “Or” does not automatically mean add, and “and” does not automatically mean multiply. Check overlap for alternatives and check the dependence of choices for stages.

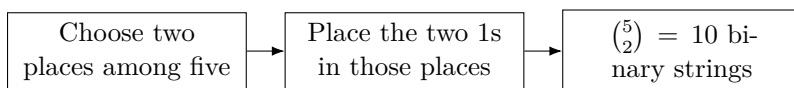
3 Factorials and binomial coefficients

The factorial $n!$ means $n(n-1)\cdots 2\cdot 1$, with $0! = 1$. It counts arrangements of n distinct objects: choose the first in n ways, the second in $n-1$, and continue until only one remains.

The binomial coefficient $\binom{n}{r}$ counts unordered selections of r distinct objects from n , where $0 \leq r \leq n$. Its formula is

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}.$$

The numerator divided by $(n-r)!$ counts ordered selections. Each unordered selection appears in $r!$ orders, so dividing by $r!$ removes precisely that duplication.



Choosing positions avoids counting the same string twice.

Worked example. Choosing three representatives from eight people gives $\binom{8}{3} = 56$. Assigning three distinct positions among the eight gives $8 \cdot 7 \cdot 6 = 336$. The factor of $3! = 6$ between the answers records the six orders of each selected triple.

These coefficients will reappear in generating functions. The coefficient of x^r in $(1+x)^n$ is $\binom{n}{r}$: from each of n factors, choose either 1 or x , and choose the x term in exactly r factors.

Practice. How many length-five binary strings contain exactly two 1s?

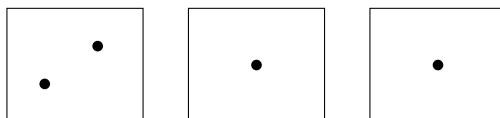
Solution. Choose the two positions that contain 1; all remaining positions must contain 0. There are $\binom{5}{2} = 10$ choices. No extra factor is needed for the zeros or the order of the ones, because positions already determine the entire string.

Common error. The formula $\binom{n}{r}$ assumes selection without repetition from distinct objects. It does not directly count unrestricted selections of repeated coin denominations. Generating functions will handle those situations by representing the permitted multiplicities explicitly.

4 The pigeonhole principle

If more than k objects are placed into k boxes, at least one box contains two or more objects. The objects are traditionally called pigeons and the categories holes, but in applications the categories may be remainders, birthdays, intervals, or possible data values.

Main idea. If every box held at most one object, all boxes together could hold at most k . Having more than k objects makes that assumption impossible. The proof is a capacity argument.



Four objects placed in three boxes

Worked example. Among 13 people, at least two were born in the same month. The people are the objects and the 12 calendar months are the boxes. We need not know their actual birthdays or which month repeats.

Practice. Prove that among any 11 integers, two have the same last decimal digit.

Solution. There are ten possible final digits, 0 through 9. Classify each integer by its residue modulo 10; this avoids ambiguity with negative-number notation. Eleven integers placed into ten residue classes force two into the same class. Their difference is therefore divisible by 10.

Common error. The principle guarantees existence, not the identity of the repeated category. It also does not require equal-sized or equally likely categories. Probability plays no role in the argument.

The main creative step is choosing useful boxes. Once every object has been assigned to exactly one of a known number of categories, the conclusion usually follows immediately.

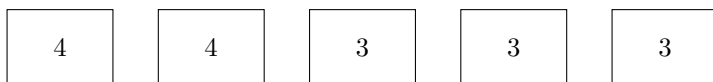
5 The generalised pigeonhole principle

If N objects are placed into k boxes (which may be empty), with $k \geq 1$, some box contains at least

$$\left\lceil \frac{N}{k} \right\rceil$$

objects. The ceiling $\lceil t \rceil$ is the smallest integer greater than or equal to the real number t . For example, $\lceil 17/5 \rceil = 4$.

To understand the formula, compare the most crowded box with the average occupancy N/k . If every box were below that average, the total would be less than N . Since occupancies are integers, at least one must reach the ceiling of the average.



17 objects in 5 boxes: some box has at least 4.

Worked example. With 25 students in six tutorial groups, some group has at least $\lceil 25/6 \rceil = 5$ students. This does not say that every group has five students or that some group has exactly five. One group might contain many more.

There is a dual observation: some box contains at most $\lfloor N/k \rfloor$ objects, where the floor is the greatest integer at most its input. This follows by comparing the least crowded box with the average.

Practice. In a set of 73 people, show that at least seven share a birth month. Could the same argument guarantee eight?

Solution. The lower bound is $\lceil 73/12 \rceil = 7$, so seven are guaranteed. Eight are not forced: distributing the people as one group of seven and eleven groups of six uses $7 + 66 = 73$ people with no month containing eight. This distribution also shows that the bound is sharp for the available information.

Common error. Round upward when bounding the maximum occupancy from below. Rounding $73/12$ down would give a true but weaker guarantee and miss the strongest conclusion supplied by the principle.

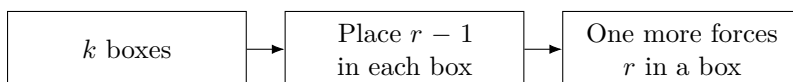
6 Threshold questions and worst-case distributions

To guarantee at least r objects in one of k boxes, the smallest sufficient total is

$$k(r - 1) + 1.$$

The reason is concrete: a distribution avoiding r in every box may place at most $r - 1$ in each, for a total of at most $k(r - 1)$. One additional object makes avoidance impossible.

This method answers “how many are needed to guarantee” questions. It begins by constructing the largest possible unsuccessful arrangement rather than guessing a successful one.



Threshold: $k(r - 1) + 1$.

Worked example. How many people guarantee that five have the same birth month? The largest arrangement with at most four per month has $12 \cdot 4 = 48$ people. The next person forces a fifth in some month, so the answer is 49.

Practice. Socks come in four colours, with at least ten of each available. How many blind selections guarantee three socks of one colour? How many guarantee three socks of a specified colour, if there is no fixed bound on the quantities of the other colours?

Solution. For any colour, at most two of each colour can be drawn without a triple. That gives eight unsuccessful draws, so nine guarantee a triple. For a specified colour, no finite bound follows from the given unbounded supply of the other colours: a selection could keep drawing those colours. The second question therefore needs additional inventory information.

Common error. A guarantee of “some category” is not a guarantee about a prescribed category. Also respect finite supplies: if a box cannot contain $r - 1$ objects because fewer are available, the worst-case calculation must use that smaller capacity.

When capacities vary, the same reasoning becomes $c_1 + \cdots + c_k + 1$, where c_i is the maximum permitted occupancy of box i while still avoiding the desired event.

7 Remainders as pigeonholes

For a positive integer m , every integer has exactly one remainder among $0, 1, \dots, m-1$ upon division by m . These m residue classes provide a natural finite set of boxes, even when the integers themselves are arbitrarily large.

Two integers with the same remainder differ by a multiple of m . Indeed, if $a = mu + r$ and $b = mv + r$, then $a - b = m(u - v)$. We write $a \equiv b \pmod{m}$ for this relationship.

remainder	remainder	remainder	remainder	remainder
0	1	2	3	4

Six integers must repeat a remainder modulo five.

Worked example. Among any $m + 1$ integers, two have a difference divisible by m . Put each integer into its remainder class modulo m . The pigeonhole principle gives a repeated class, and the displayed calculation converts the shared category into the desired divisibility statement.

Practice. Prove that from any six integers, two can be selected whose difference is divisible by 5. Does the same argument prove that two have a sum divisible by 5?

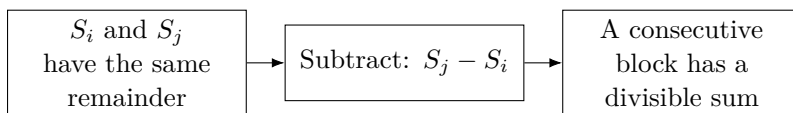
Solution. The difference claim follows from six objects and five remainder classes. The same-class argument does not establish the sum claim: two numbers with remainder 1 have sum with remainder 2. In fact, the sum claim is false without further assumptions; six integers all congruent to 1 modulo 5 have no pair whose sum is divisible by 5.

Common error. Select categories according to the desired algebraic conclusion. Equal remainders control differences. Complementary remainders may help control sums, but they require a different grouping and often different hypotheses.

The use of residues is a compression of information. Instead of tracking the entire integer, retain only the part that matters for divisibility by the chosen modulus. A good pigeonhole proof discards irrelevant detail while preserving the feature needed in the conclusion.

8 Partial sums and a divisible consecutive block

Pigeonhole arguments can act on derived objects rather than the original data. Partial sums are especially useful when a conclusion concerns a consecutive block of terms, because a block sum is the difference of two partial sums.



Use partial sums to turn a block sum into a difference.

Worked example. Given any m integers $a_1, \dots, a_m$, there is a nonempty consecutive block whose sum is divisible by m . Define

$$S_0 = 0, \quad S_j = a_1 + \dots + a_j \quad (1 \leq j \leq m).$$

There are $m + 1$ partial sums and only m remainder classes modulo m . Therefore S_i and S_j have the same remainder for some $i < j$. Their difference is

$$S_j - S_i = a_{i+1} + \dots + a_j,$$

which is divisible by m . Since $i < j$, the block is nonempty.

Practice. Find such a block in the list 2, 5, 1, 4 with $m = 4$, using partial sums.

Solution. The partial sums are 0, 2, 7, 8, 12, whose remainders modulo 4 are 0, 2, 3, 0, 0. The equal remainders of S_0 and S_3 give the block $2 + 5 + 1 = 8$. The equal remainders of S_3 and S_4 give the one-term block 4. Either block is a valid answer; the existence theorem does not claim uniqueness.

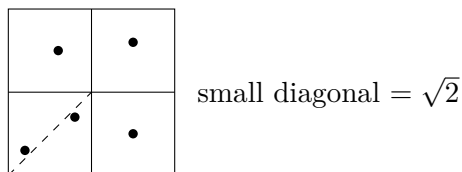
Common error. Include $S_0 = 0$. It allows a block starting with the first term to appear as a difference just like every other block, and it gives exactly the extra object required by the pigeonhole principle.

The proof works with negative integers as well as positive ones. Positivity is not needed because the argument concerns remainders and differences, not monotonicity of the partial sums.

9 Geometric pigeonhole arguments

Geometry supplies boxes by partitioning a region. If two points must fall in the same small region, a bound on that region's diameter gives a bound on their distance. The *diameter* is the largest possible distance between two points of the region.

Worked example. Place five points in a square of side length 2. Divide the square into four unit squares. At least two points lie in the same small square, and their distance is at most its diagonal, $\sqrt{2}$.



Boundary points need a consistent assignment rule, such as assigning an internal dividing-line point to the square on its right or above. The essential requirement is that every point belongs to exactly one box for counting purposes. Every assigned square still has diagonal $\sqrt{2}$.

Practice. Show that among ten points in a unit square, two have distance at most $\sqrt{2}/3$.

Solution. Divide the square into a 3 by 3 grid. There are nine squares of side $1/3$, so ten points force two into one small square. Its diagonal is $\sqrt{(1/3)^2 + (1/3)^2} = \sqrt{2}/3$, giving the required bound.

Common error. The diagonal bounds Euclidean distance; the side length generally does not. Two points near opposite corners of the same small square can be farther apart than its side length.

These arguments provide guaranteed bounds. A partition may not give the best possible geometric constant, so do not describe a bound as optimal unless sharpness has also been established.

10 Designing boxes from a hidden structure

Sometimes the right categories are not obvious from the statement. A useful strategy is to remove one feature of each object until a simpler representative remains. If equal representatives imply the desired relation, they can serve as boxes.



These numbers share odd part 1;
each divides the later ones.

Worked example. Select $n + 1$ distinct integers from $\{1, 2, \dots, 2n\}$. Then one selected integer divides another. Every positive integer can be written uniquely as $2^k u$, where u is odd and $k \geq 0$: repeatedly divide by 2 until an odd number remains. Use this odd part u as the box label.

There are only n possible odd labels among $1, 3, \dots, 2n - 1$. Two selected numbers therefore have the same odd part, say $2^a u$ and $2^b u$. Since the numbers are distinct, $a \neq b$. If $a < b$, then $2^b u = 2^{b-a}(2^a u)$, so the smaller power-of-two multiple divides the larger.

Practice. Among $2, 3, 4, 5, 6, 7$ selected from $\{1, \dots, 10\}$, group the numbers by odd part and identify a dividing pair.

Solution. The odd parts are 1, 3, 1, 5, 3, 7. Thus 2 and 4 share odd part 1, and 3 and 6 share odd part 3. We obtain the pairs $2 \mid 4$ and $3 \mid 6$. The theorem promises at least one, and this selection happens to supply two.

Common error. Distinctness is part of this theorem's useful conclusion. If repeated selections were allowed, identical numbers would trivially divide each other, changing the nature of the problem. Always keep the original object definition visible while constructing categories.

The proof illustrates a general tactic: identify an invariant, meaning a feature that stays unchanged under a relevant transformation. Here multiplying by 2 leaves the odd part unchanged, and shared odd parts reveal a divisibility chain.

11 Sharp bounds and the limitations of counting guarantees

A lower bound is *sharp* when an example achieves it, showing that it cannot be improved under the given assumptions. A guarantee proof and a sharpness example serve different purposes: the proof covers every arrangement, while the example blocks a stronger universal claim.

Worked example. With 17 objects in five boxes, at least four occupy some box. The distribution $(4, 4, 3, 3, 3)$ has total 17 and no occupancy above four. Therefore a guarantee of five would be false, and the bound four is sharp.

In the divisibility theorem from the previous section, $n + 1$ chosen integers suffice. The threshold cannot be reduced to n : the set $\{n + 1, n + 2, \dots, 2n\}$ has n elements and no distinct pair with one dividing the other. Any larger multiple of one of these elements exceeds $2n$.

Practice. Find the smallest number of people that guarantees six share a birth month, and justify that the number is minimal.

Solution. At most five per month gives $12 \cdot 5 = 60$ people without six in a month. Thus 61 suffice. The distribution of exactly five people in each month shows that 60 do not suffice. Both parts are needed to justify the word “smallest.”

Common error. An argument that 61 works establishes sufficiency, not minimality. A minimal threshold requires an unsuccessful arrangement immediately below the proposed threshold, or another lower-bound argument excluding every smaller candidate.

Pigeonhole reasoning says little about where the crowded box is or how other boxes are populated. It remains powerful precisely because it needs so little information: a total count and a limited number of categories can force a structural conclusion.

12 Pigeonhole practice with complete solutions

Practice.

1. Among 31 integers, prove that four have the same remainder modulo 10.
2. How many draws from five categories guarantee four objects in some category?
3. Show that among seven real numbers in $[0, 1]$, two differ by at most $1/6$.

Solution. 1. There are ten possible remainders. The generalised principle gives a box with at least $\lceil 31/10 \rceil = 4$ members. The conclusion is about a shared remainder, and hence all pairwise differences within that group are divisible by 10.

2. The largest distribution avoiding four has three objects in each of five categories, for 15 objects. Therefore 16 guarantee four. The arrangement $(3, 3, 3, 3, 3)$ shows minimality if supplies allow it.

3. Divide $[0, 1]$ into six intervals of length $1/6$, using half-open intervals except for the final one, which includes 1. Every point is assigned to exactly one interval. Seven numbers force two into one interval, and the difference of two numbers in such an interval has absolute value at most $1/6$.

A second solution to 3. Sort the numbers as $x_1 \leq \dots \leq x_7$. The six adjacent gaps sum to $x_7 - x_1 \leq 1$. If all gaps exceeded $1/6$, their sum would exceed 1, a contradiction. Hence some adjacent pair has the required distance. This is the same capacity idea expressed through gaps rather than interval boxes.

Common error. In a proof about real numbers, distinguish the signed difference $a - b$ from the distance $|a - b|$. A statement that two numbers “differ by at most” a bound normally concerns the absolute value.

Before moving to recurrences, explain the common structure of all three solutions: identify finitely many places in which the relevant objects or total size can be distributed, then show that uniform avoidance would violate the total.

Practice bank: Pigeonhole questions

Attempt these questions before reading the answers on the following page. For a calculation or proof, write the method and the essential working as well as the final result.

1. How many people guarantee that three share a birth month?
2. When 41 objects enter eight boxes, how many objects must some box contain?
3. How many draws from five colours guarantee a pair of one colour?
4. How many draws from five colours guarantee four of one colour?
5. Why do twelve integers guarantee two with the same remainder modulo eleven?
6. Find a consecutive block in $1, 2, 4$ whose sum is divisible by three.
7. Nine points lie in a square of side two. Give a guaranteed bound on the distance of some pair using four equal squares.
8. Give an arrangement showing that 20 objects in four boxes need not put six in a box.

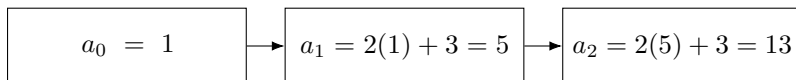
Answers: Pigeonhole questions

Compare the reasoning as well as the final answer. If your answer differs, return to the first step where the methods separate.

1. At most two in each of twelve months gives 24 without a triple. Therefore 25 guarantee three.
2. Some box contains at least $\lceil 41/8 \rceil = 6$ objects.
3. Six. Five draws could give one of each colour; the next must repeat a colour.
4. $5(4-1) + 1 = 16$, assuming the supplies allow three of each colour in the avoiding arrangement.
5. There are only eleven remainder classes. Two integers share a class, so their difference is divisible by eleven.
6. The block $1 + 2 = 3$ works. The partial sums 0, 1, 3, 7 contain two with remainder zero modulo three.
7. Two points share a unit square, so their distance is at most its diagonal $\sqrt{2}$. This partition does not claim the best possible bound.
8. Use occupancies (5, 5, 5, 5). This shows why 21, rather than 20, is needed to guarantee six.

13 Reading a recurrence question

A recurrence tells you how to calculate a new term from earlier terms. In $a_n = 2a_{n-1} + 3$, a_n means the current term and a_{n-1} means the previous term. The instruction is simply: double the previous term, then add three.



Each subscript identifies a position in the sequence.

Worked example. If $a_0 = 1$, put $n = 1$ into the given rule:

$$a_1 = 2a_0 + 3 = 5.$$

Next put $n = 2$: $a_2 = 2a_1 + 3 = 13$. Continuing gives 1, 5, 13, 29, ... This finds successive terms. To find a formula for any term, use the auxiliary-equation method developed below.

Identify the type before solving.

1. Find the oldest term mentioned. A rule using a_{n-1} is first order; one using a_{n-2} is second order.
2. Look for an extra term that does not contain an a . If there is none, the recurrence is homogeneous. An extra 3, n , or 2^n makes it nonhomogeneous.
3. Check the initial values. A usual second-order question gives two, such as a_0 and a_1 .

Linear with constant coefficients means the sequence terms are not squared or multiplied together, and their multipliers are fixed numbers. The main methods in these notes apply to this type.

Practice. Classify $b_n = 3b_{n-1} - 2b_{n-2} + 5$.

Solution. It is second order because it uses b_{n-2} . It is nonhomogeneous because of the extra 5. Its fixed multipliers are 3 and -2 . Solve its homogeneous part with an auxiliary equation, then find a response to the extra 5.

Common error. An initial value such as $a_0 = 1$ is starting information. It is not an additional term to put into the recurrence.

14 Building a recurrence from a description

Before solving a recurrence, justify that it models the intended process. Define the quantity being counted, divide the possible objects by a meaningful final step, and show that the cases are disjoint and exhaustive.

Worked example. Let t_n count ways to tile a $1 \times n$ strip using 1×1 squares and 1×2 dominoes. A tiling ends either in a square, leaving a tiling of length $n - 1$, or in a domino, leaving one of length $n - 2$. These cases are disjoint and cover every tiling. Therefore

$$t_n = t_{n-1} + t_{n-2} \quad (n \geq 2).$$

The initial values are $t_0 = 1$ and $t_1 = 1$. There is one empty tiling of an empty strip; this convention lets a domino by itself be counted through the t_0 contribution when $n = 2$.

length $n - 1$	1
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length $n - 2$	2
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Practice. Find t_2, t_3, t_4 and explain why the recurrence does not multiply the two preceding counts.

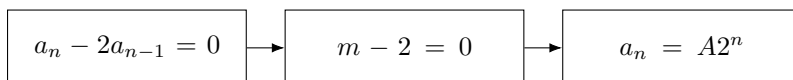
Solution. The values are 2, 3, 5. We add because a completed tiling belongs to one of two alternative final-piece cases. We are not independently choosing both a length- $(n - 1)$ tiling and a length- $(n - 2)$ tiling to make one object, so the product principle does not apply.

Common error. A recurrence guessed from several initial values is not yet a justified model. Explain the structural correspondence that makes it hold for every permitted n .

15 First-order auxiliary equations

For a first-order homogeneous recurrence, the auxiliary equation is especially short. Use the letter m for its unknown. The letter m is an algebraic tool; it is not the index n .

Recipe. Move the sequence terms to the left. Replace a_n by m and a_{n-1} by 1. Solve the resulting equation. If the root is $m = r$, write $a_n = Ar^n$, then find A from the given initial value.



Form the auxiliary equation before using the initial value.

Worked example. Solve $a_n = 2a_{n-1}$, given $a_0 = 3$.

1. Rearrange: $a_n - 2a_{n-1} = 0$.
2. Form the auxiliary equation: $m - 2 = 0$.
3. The root is $m = 2$, so write $a_n = A2^n$.
4. Put $n = 0$: $3 = A2^0 = A$. Therefore $a_n = 3 \cdot 2^n$.

As a quick check, the formula gives $a_1 = 6$, which is twice the starting value 3.

Two different substitutions. Replacing the sequence terms by powers of m forms the auxiliary equation. Substituting $n = 0$ or $n = 1$ comes later, when finding constants from initial values. These are separate steps.

Practice. Solve $b_n = -2b_{n-1}$ with $b_0 = 4$.

Solution. Rearrangement gives $b_n + 2b_{n-1} = 0$, so the auxiliary equation is $m + 2 = 0$. Its root is -2 . Write $b_n = A(-2)^n$ and use $b_0 = 4$ to get $A = 4$. Thus $b_n = 4(-2)^n$. The negative root makes the signs alternate.

Common error. If the starting value is a_1 rather than a_0 , substitute $n = 1$ when finding A . Do not silently treat it as a value at index zero.

16 The basic example: doubling and adding a constant

Consider $a_n = 2a_{n-1} + c$, where c is a fixed number. The extra constant means we need two pieces: the complementary function, abbreviated CF, and a particular solution, abbreviated PI. The CF solves the rule with no extra term. The PI accounts for the extra term.

Step 1: find the CF. Temporarily leave out c . The rule becomes $a_n - 2a_{n-1} = 0$, so its auxiliary equation is $m - 2 = 0$. The root is 2, giving $\text{CF}_n = A2^n$.

Step 2: find the PI. Because the extra term is constant, try a constant K . Put this into the original rule:

$$K = 2K + c, \quad K = -c.$$

Step 3: combine and use the initial value.

$$a_n = A2^n - c.$$

If $a_0 = d$, substituting $n = 0$ gives $d = A - c$, hence $A = d + c$. Therefore

$$a_n = (d + c)2^n - c.$$

Worked example. For $a_n = 2a_{n-1} + 3$ with $a_0 = 1$, the answer is $a_n = 4 \cdot 2^n - 3$. At $n = 1$ this gives 5, agreeing with $2(1) + 3$.

Practice. Solve $b_n = 2b_{n-1} + 5$, given $b_0 = 2$.

Solution. The auxiliary equation is $m - 2 = 0$, so the CF is $A2^n$. A constant trial gives $K = 2K + 5$, so the PI is -5 . Now $b_n = A2^n - 5$. At $n = 0$, $2 = A - 5$, giving $A = 7$. Hence $b_n = 7 \cdot 2^n - 5$.

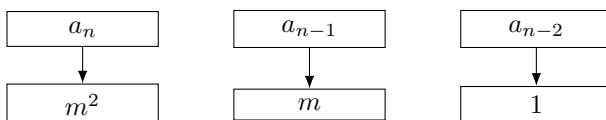
Common error. Include the extra constant when finding the PI. Leave it out only while finding the CF.

17 Forming a second-order auxiliary equation

For a second-order recurrence, use this replacement pattern after moving all sequence terms to the left:

$$a_n \longrightarrow m^2, \quad a_{n-1} \longrightarrow m, \quad a_{n-2} \longrightarrow 1.$$

Keep every numerical coefficient and its sign. This gives the auxiliary equation, also called the characteristic equation.



The second-order replacement pattern.

Worked example. Form the auxiliary equation for $a_n = 5a_{n-1} - 6a_{n-2}$. First rearrange:

$$a_n - 5a_{n-1} + 6a_{n-2} = 0.$$

Now replace the three sequence terms:

$$m^2 - 5m + 6 = 0.$$

Factor it as $(m - 2)(m - 3) = 0$. The roots are 2 and 3.

Because the roots are different, write

$$a_n = A2^n + B3^n.$$

The letters A, B are unknown constants. Two initial values will give two equations for them.

Why the replacement works, briefly. A power such as m^n changes by a fixed factor when the index changes. That matches the fixed multipliers in the recurrence. For an ordinary solving question, the important working is the auxiliary equation and the steps that follow it.

Practice. Form and factor the auxiliary equation for $b_n = 4b_{n-1} + 5b_{n-2}$.

Solution. Rearrange to $b_n - 4b_{n-1} - 5b_{n-2} = 0$. The auxiliary equation is $m^2 - 4m - 5 = 0$, or $(m - 5)(m + 1) = 0$. The roots are 5, -1 , so the answer form is $b_n = A5^n + B(-1)^n$.

Common error. The $+5b_{n-2}$ on the right becomes $-5b_{n-2}$ on the left. Copying its old sign would give the wrong auxiliary equation.

18 Different roots: a complete exam solution

When the auxiliary equation has two different roots r, s , use

$$a_n = Ar^n + Bs^n.$$

Then put in the given starting indices to find A, B . Do not try to guess these constants from the roots.

Worked example. Solve $a_n = 5a_{n-1} - 6a_{n-2}$ with $a_0 = 2, a_1 = 5$.

1. Auxiliary equation.

$$m^2 - 5m + 6 = 0 \implies (m - 2)(m - 3) = 0.$$

2. General answer. The roots are 2, 3, so $a_n = A2^n + B3^n$.

3. Initial values. At $n = 0$, $A + B = 2$, since both powers equal 1. At $n = 1$, $2A + 3B = 5$.

4. Solve the two equations. Double $A + B = 2$ to get $2A + 2B = 4$. Subtract this from $2A + 3B = 5$. Therefore $B = 1$, and then $A = 1$.

5. Final answer. $a_n = 2^n + 3^n$. As a check, $a_2 = 4 + 9 = 13$, while the recurrence gives $5(5) - 6(2) = 13$.

Practice. Solve $b_n = 3b_{n-1} - 2b_{n-2}$ with $b_0 = 0, b_1 = 1$.

Solution. The auxiliary equation is $(m - 1)(m - 2) = 0$. Write $b_n = A + B2^n$. The initial values give $A + B = 0$ and $A + 2B = 1$. Subtract to get $B = 1$, then $A = -1$. Hence $b_n = 2^n - 1$.

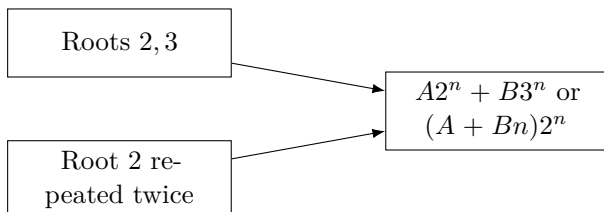
Common error. A root equal to 1 contributes $A1^n = A$. Keep that constant term; it is one of the two pieces needed to fit the initial values.

19 Repeated roots: the one change to remember

If the auxiliary equation has the same root r twice, the answer form is

$$a_n = (A + Bn)r^n.$$

The extra factor n is essential. Writing $Ar^n + Br^n$ would combine into just one constant times r^n , leaving too little freedom to use two initial values.



Repeated roots change the answer form.

Worked example. Solve $a_n = 4a_{n-1} - 4a_{n-2}$, with $a_0 = 1, a_1 = 6$.

1. Rearrange and form the auxiliary equation: $m^2 - 4m + 4 = 0$.
2. Factor: $(m - 2)^2 = 0$. The root 2 is repeated.
3. Write $a_n = (A + Bn)2^n$.
4. Put $n = 0$: $1 = A$.
5. Put $n = 1$: $6 = (1 + B)2$, so $B = 2$.

Thus $a_n = (1 + 2n)2^n$. The next term is 20; the recurrence also gives $4(6) - 4(1) = 20$.

Practice. Solve $b_n = 2b_{n-1} - b_{n-2}$ with $b_0 = 3, b_1 = 8$.

Solution. The auxiliary equation is $m^2 - 2m + 1 = (m - 1)^2 = 0$. Therefore $b_n = (A + Bn)1^n = A + Bn$. The initial values give $A = 3$ and $3 + B = 8$. Hence $b_n = 3 + 5n$.

Quick comparison. Different roots give $Ar^n + Bs^n$. A repeated root gives $(A + Bn)r^n$. Choose the form after factoring, not before.

Common error. For a root repeated three times, use $(A + Bn + Cn^2)r^n$. The highest power of n is one less than the number of repetitions.

20 When the auxiliary roots contain an imaginary number

Most introductory examples factor into real roots. If the quadratic formula gives complex roots, the auxiliary-equation steps still apply. The symbol i means $\sqrt{-1}$, so $i^2 = -1$.

Worked example. Solve $a_n = -a_{n-2}$ with $a_0 = 1, a_1 = 0$.

Rearrange to $a_n + a_{n-2} = 0$. Replace the terms by m^2 and 1:

$$m^2 + 1 = 0, \quad m = i, -i.$$

The roots are different, so write $a_n = Ai^n + B(-i)^n$. At $n = 0$, $A + B = 1$. At $n = 1$, $Ai - Bi = 0$, so $A = B$. Therefore

$$a_n = \frac{i^n + (-i)^n}{2}.$$

The values are real: $1, 0, -1, 0, 1, 0, \dots$. You can also obtain them directly from the recurrence.

If a real-valued trigonometric form is required. For roots $\rho(\cos \theta \pm i \sin \theta)$, use

$$a_n = \rho^n (C \cos(n\theta) + D \sin(n\theta)).$$

Here ρ is the magnitude of a root and θ its angle. For the example, $\rho = 1, \theta = \pi/2$, giving $a_n = \cos(n\pi/2)$.

Practice. Find the first six terms when the same recurrence has $b_0 = 0, b_1 = 2$.

Solution. Each term is the negative of the term two places earlier. Thus the terms are $0, 2, 0, -2, 0, 2$. A real formula is $b_n = 2 \sin(n\pi/2)$.

Common error. Do not abandon the problem when a root contains i . Use the same initial-value substitutions. If the question asks only for early terms, direct substitution into the recurrence may be enough.

21 Third-order auxiliary equations

For a third-order recurrence, replace the sequence terms in descending order:

$$a_n \rightarrow m^3, \quad a_{n-1} \rightarrow m^2, \quad a_{n-2} \rightarrow m, \quad a_{n-3} \rightarrow 1.$$

The method is the same as for second order. A usual third-order question supplies three initial values, giving three equations for the constants.

Worked example. Consider $a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}$. Its auxiliary equation is

$$m^3 - 6m^2 + 11m - 6 = 0.$$

Factor it as $(m-1)(m-2)(m-3) = 0$. The roots are 1, 2, 3, so the solution form is $a_n = A + B2^n + C3^n$.

If $a_0 = 3, a_1 = 6, a_2 = 14$, substitution gives

$$A + B + C = 3,$$

$$A + 2B + 3C = 6,$$

$$A + 4B + 9C = 14.$$

Subtract consecutive equations: $B + 2C = 3$ and $2B + 6C = 8$. Subtract twice the first of these from the second to get $2C = 2$. Hence $C = 1, B = 1, A = 1$, and $a_n = 1 + 2^n + 3^n$.

Practice. An auxiliary equation is $(m-2)^2(m+1) = 0$. Write the answer form.

Solution. The repeated root 2 contributes $(A + Bn)2^n$. The single root -1 contributes $C(-1)^n$. Therefore $a_n = (A + Bn)2^n + C(-1)^n$.

Common error. If the recurrence has no a_{n-2} term, the coefficient of m is zero. Do not shift the other powers to close the gap.

22 Using initial values without index mistakes

After solving the auxiliary equation, your answer still contains constants such as A, B . Initial values determine these constants. The most common substitutions are $n = 0$ and $n = 1$, but always use the indices actually given.

Worked example. Suppose the answer form is $a_n = A2^n + B3^n$ and the question gives $a_1 = 5, a_2 = 13$. Put $n = 1$ and $n = 2$:

$$2A + 3B = 5, \quad 4A + 9B = 13.$$

Double the first equation to get $4A + 6B = 10$. Subtract it from the second: $3B = 3$, so $B = 1$. Now $2A + 3 = 5$, giving $A = 1$. The answer is $a_n = 2^n + 3^n$.

For a nonhomogeneous question, substitute the initial values into the complete answer, CF plus PI. If $a_n = A2^n - 3$ and $a_0 = 1$, then $1 = A - 3$, not $1 = A$.

Quick check after solving.

1. Put the given starting indices into your final formula.
2. Compute one additional term using the recurrence.
3. Check that the formula gives the same term.

These checks often catch sign and arithmetic errors. In a proof question, also show that the formula satisfies the recurrence in general.

Practice. The form is $b_n = (A + Bn)2^n$, with $b_0 = 2, b_1 = 8$. Find A, B .

Solution. At $n = 0$, $A = 2$. At $n = 1$, $8 = 2(A + B) = 2(2 + B)$. Thus $B = 2$, and $b_n = (2 + 2n)2^n$.

Common error. In $(A + Bn)2^n$, substituting $n = 1$ changes both Bn and 2^n . Do not substitute into only one of them.

Practice bank: Auxiliary equations and initial values

Attempt these questions before reading the answers on the following page. For a calculation or proof, write the method and the essential working as well as the final result.

1. Form the auxiliary equation for $a_n = 7a_{n-1}$.
2. Solve that recurrence with $a_0 = 2$.
3. Form and factor the auxiliary equation for $b_n = 4b_{n-1} - 3b_{n-2}$.
4. Solve the previous recurrence with $b_0 = 2, b_1 = 4$.
5. Write the answer form for $(m - 4)^2 = 0$.
6. Find A, B in $(A + Bn)3^n$ if $a_0 = 1, a_1 = 6$.
7. Write the answer form for auxiliary roots $2, -3$.
8. For a third-order recurrence, what replaces a_{n-2} when forming the auxiliary equation?

Answers: Auxiliary equations and initial values

Compare the reasoning as well as the final answer. If your answer differs, return to the first step where the methods separate.

1. Rearrange to $a_n - 7a_{n-1} = 0$ and replace the terms by $m, 1$. The equation is $m - 7 = 0$.
2. The root is 7, so $a_n = A7^n$. At zero, $A = 2$, giving $a_n = 2 \cdot 7^n$.
3. The equation is $m^2 - 4m + 3 = 0 = (m - 1)(m - 3)$. The roots are 1, 3.
4. Write $b_n = A + B3^n$. Then $A + B = 2$ and $A + 3B = 4$. Thus $A = B = 1$, and $b_n = 1 + 3^n$.
5. The double root is 4, so use $(A + Bn)4^n$.
6. The first condition gives $A = 1$. The second gives $3(1 + B) = 6$, so $B = 1$.
7. Use $A2^n + B(-3)^n$. The second contribution alternates in sign.
8. Use m . The complete pattern is $m^3, m^2, m, 1$ for $a_n, a_{n-1}, a_{n-2}, a_{n-3}$.

23 Nonhomogeneous questions: CF plus PI

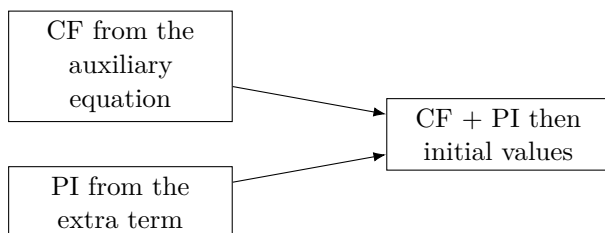
If the recurrence has an extra term such as 5, n , or 3^n , use

$$\boxed{\text{complete answer} = \text{CF} + \text{PI.}}$$

The CF is found from the auxiliary equation after temporarily setting the extra term to zero. The PI is a trial expression chosen to match the type of extra term.

Extra term	First PI trial
Constant, such as 5	K
Linear, such as $2n + 1$	$Kn + L$
Quadratic, such as n^2	$Kn^2 + Ln + M$
Exponential, such as 3^n	$K3^n$

Here K, L, M are constants to calculate by substitution. Sometimes a trial must be multiplied by n or n^2 because it overlaps the CF; the next sections explain when.



Use initial values in the complete expression.

Worked example. Solve $a_n = 3a_{n-1} + 4$ with $a_0 = 2$.

1. Auxiliary equation: $m - 3 = 0$. Thus $\text{CF} = A3^n$.
2. The extra term is constant, so try $\text{PI} = K$.
3. Substitute into the original recurrence: $K = 3K + 4$, giving $K = -2$.
4. Combine: $a_n = A3^n - 2$.
5. Use $a_0 = 2$: $2 = A - 2$, so $A = 4$.

The answer is $a_n = 4 \cdot 3^n - 2$.

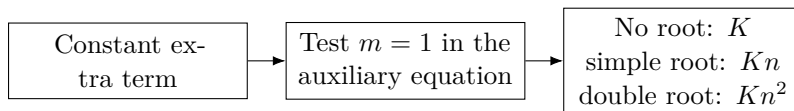
Practice. What trial would you first use for the extra term $7n + 4$?

Solution. Use $Kn + L$. A constant trial cannot generally account for the changing n term. Substitute the full trial, including its constant part.

Common error. Find the PI before using initial values. The PI is not a separate sequence that needs to match the starting data by itself.

24 Constant extra terms: when to try a constant

Start with a constant PI when the extra term is a constant. There is one important exception: if 1 is a root of the auxiliary equation, a constant already appears in the CF. Then try a term involving n instead.



Choose the trial before finding its coefficient.

Worked example. Solve $a_n = 2a_{n-1} + 3$, $a_0 = 1$. The auxiliary equation $m - 2 = 0$ has no root 1. Try PI = K :

$$K = 2K + 3, \quad K = -3.$$

Thus $a_n = A2^n - 3$. Using $a_0 = 1$ gives $A = 4$, so $a_n = 4 \cdot 2^n - 3$.

The exception in a simple example. For $b_n = b_{n-1} + 3$, the auxiliary equation is $m - 1 = 0$, so the CF is A . A constant PI would give $K = K + 3$, which cannot work. Try PI = Kn :

$$Kn = K(n - 1) + 3.$$

Cancel Kn to obtain $K = 3$. Therefore $b_n = A + 3n$. If $b_0 = 2$, the answer is $b_n = 2 + 3n$.

Practice. Find a PI for $c_n = 2c_{n-1} + c_{n-2} + 6$.

Solution. The auxiliary equation is $m^2 - 2m - 1 = 0$. Substituting $m = 1$ does not give zero, so a constant trial is suitable. Set $c_n = c_{n-1} = c_{n-2} = K$. Then $K = 2K + K + 6$, giving $K = -3$.

Common error. Substituting 1 here means testing $m = 1$ in the auxiliary polynomial to decide the PI trial. It is different from substituting $n = 1$ to use an initial value.

If 1 is a double root, multiply the constant trial by n^2 . Keep this adjustment rule beside the trial table.

25 An extra term involving n

For an extra term that is linear in n , try $\text{PI}_n = Kn + L$. Use two letters even if the extra term is just n : shifting the index creates constants that must also be matched.

Worked example. Solve $a_n = 2a_{n-1} + n$, given $a_0 = 0$.

CF. The auxiliary equation is $m - 2 = 0$, so $\text{CF} = A2^n$.

PI. Try $Kn + L$. The previous term of this trial is $K(n - 1) + L$. Substitute:

$$Kn + L = 2(K(n - 1) + L) + n.$$

Expanding the right side gives $(2K + 1)n + (2L - 2K)$. Match the coefficients of n and the constants:

$$K = 2K + 1, \quad L = 2L - 2K.$$

Hence $K = -1$ and $L = -2$. The PI is $-n - 2$.

Initial value. The complete answer is $a_n = A2^n - n - 2$. At $n = 0$, $0 = A - 2$, so

$$a_n = 2^{n+1} - n - 2.$$

Practice. What trial should you use for $3n^2 - 5n + 7$ if 1 is not an auxiliary root?

Solution. Use $Kn^2 + Ln + M$. Include the linear and constant terms as well as the quadratic term. Substitute $n - 1$ or $n - 2$ throughout the trial wherever the recurrence needs an earlier term.

Common error. If 1 is an auxiliary root, multiply the whole polynomial trial by the needed power of n . For a simple root use $n(Kn + L)$; for a double root use $n^2(Kn + L)$.

26 An extra term such as 3 to the power n

For an extra term cb^n , first try $\text{PI} = Kb^n$. Before substituting, check whether b is a root of the auxiliary equation. If it is not, this trial is the usual choice.

Worked example. Solve $a_n = 2a_{n-1} + 3^n$, given $a_0 = 0$.

The auxiliary equation is $m - 2 = 0$, so the CF is $A2^n$. The extra term has base 3, which is not the auxiliary root 2. Try $\text{PI} = K3^n$. Its previous term is $K3^{n-1}$. Substitution gives

$$K3^n = 2K3^{n-1} + 3^n.$$

Divide every term by 3^{n-1} :

$$3K = 2K + 3, \quad K = 3.$$

Therefore $a_n = A2^n + 3 \cdot 3^n$. At $n = 0$, $0 = A + 3$, giving

$$a_n = 3(3^n - 2^n).$$

At $n = 1$, the answer is 3, matching the given recurrence.

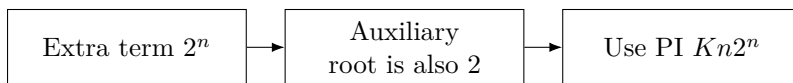
Practice. Find a PI for $b_n = 4b_{n-1} + 2^n$.

Solution. The auxiliary root is 4, so try $K2^n$. Substitute and divide by 2^{n-1} to get $2K = 4K + 2$. Hence $K = -1$, so $\text{PI} = -2^n$. The complete form would be $b_n = A4^n - 2^n$.

Common error. Check the base, not the multiplier in front. For an extra term $7 \cdot 3^n$, test $m = 3$ in the auxiliary equation. The 7 affects the calculation of K , not the choice of base.

27 When to multiply the PI trial by n

If the extra term has a base that is already an auxiliary root, the first trial overlaps the CF. Multiply that trial by n for a simple root, or by n^2 for a double root. This situation is sometimes called resonance; the adjustment rule is what you need for solving questions.



One matching root needs one extra factor of n .

Worked example. Solve $a_n = 2a_{n-1} + 2^n$, with $a_0 = 1$.

The auxiliary equation is $m - 2 = 0$. The CF is $A2^n$. A trial $K2^n$ would overlap that CF, so use

$$\text{PI}_n = Kn2^n.$$

Its previous term is $K(n-1)2^{n-1}$. Substitute into the original rule:

$$Kn2^n = 2K(n-1)2^{n-1} + 2^n.$$

Divide by 2^n : $Kn = K(n-1) + 1$. This gives $K = 1$. Thus

$$a_n = A2^n + n2^n = (A+n)2^n.$$

At $n = 0$, $A = 1$, and the answer is $(n+1)2^n$.

Practice. Find a PI for $b_n = 3b_{n-1} + 5 \cdot 3^n$.

Solution. The auxiliary root 3 matches the extra term's base. Use $Kn3^n$. Substitution and division by 3^n give $Kn = K(n-1) + 5$, so $K = 5$. The PI is $5n3^n$.

Memory aid. No match: use the ordinary trial. One matching root: multiply by n . A double matching root: multiply by n^2 .

Common error. Only the PI trial gets this adjustment. Write the CF from the auxiliary roots as usual, then add the calculated PI.

28 A double matching root: a worked question

Suppose the extra term is a constant and the auxiliary equation has a double root at 1. The ordinary trial is a constant K , but the double match means the correct trial is Kn^2 .

Worked example. Solve

$$a_n = 2a_{n-1} - a_{n-2} + 2, \quad a_0 = 0, \quad a_1 = 1.$$

1. Find the CF. The auxiliary equation is $m^2 - 2m + 1 = (m - 1)^2 = 0$. The repeated root gives CF = $A + Bn$.

2. Choose the PI. Use Kn^2 . Its earlier terms are $K(n - 1)^2$ and $K(n - 2)^2$.

3. Substitute.

$$Kn^2 - 2K(n - 1)^2 + K(n - 2)^2 = 2.$$

Expand $(n - 1)^2 = n^2 - 2n + 1$ and $(n - 2)^2 = n^2 - 4n + 4$. The left side simplifies to $2K$, so $K = 1$.

4. Use the starting values. Now $a_n = A + Bn + n^2$. At $n = 0$, $A = 0$. At $n = 1$, $1 = B + 1$, giving $B = 0$. Therefore $a_n = n^2$.

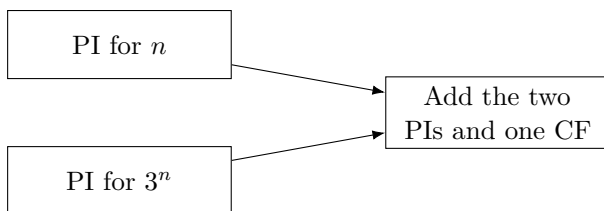
Practice. Find a PI if the extra term in this recurrence is 6 instead of 2.

Solution. The trial remains Kn^2 . Substitution again gives $2K$ on the left, but now $2K = 6$. Thus $K = 3$ and PI = $3n^2$.

Common error. A trial Kn would also overlap the CF $A + Bn$. The double root means you need n^2 , not just n .

29 Two extra terms: find two PIs and add

If the extra term is a sum, such as $n + 3^n$, handle its parts separately. Find a PI for n , find another PI for 3^n , then add them. Add the CF once, and finally use the initial values.



Treat the two extra terms separately.

Worked example. Solve $a_n = 2a_{n-1} + n + 3^n$, with $a_0 = 0$.

The auxiliary equation is $m - 2 = 0$, giving CF = $A2^n$.

For the extra term n , try $Kn + L$. As in the earlier worked question, substitution gives $K = -1, L = -2$, so the first PI is $-n - 2$.

For 3^n , try $M3^n$. Substitution gives $3M = 2M + 3$, hence $M = 3$. The second PI is $3 \cdot 3^n$.

Combine the pieces:

$$a_n = A2^n - n - 2 + 3 \cdot 3^n.$$

At $n = 0$, $0 = A - 2 + 3$, so $A = -1$. Therefore

$$a_n = -2^n - n - 2 + 3^{n+1}.$$

The next value is 4 by both the formula and the recurrence.

Practice. Choose the PI trial for

$$b_n - 3b_{n-1} + 2b_{n-2} = n + 2^n.$$

Solution. The auxiliary roots are 1 and 2. For n , the usual trial $Kn + L$ must be multiplied by n , because 1 is a simple root. For 2^n , the trial $M2^n$ must also be multiplied by n , because 2 is a simple root. Thus use $n(Kn + L) + Mn2^n$.

Common error. Check each extra term separately. One part may need an extra factor n while another part does not.

30 The recurrence answer format to practise

For a standard solving question, organise your answer under five short steps:

1. Write and solve the auxiliary equation.
2. Write the CF using the roots.
3. Find the PI if there is an extra term.
4. Add CF and PI.
5. Substitute the initial values and state the final formula.

Worked example. Solve $a_n = 3a_{n-1} - 2a_{n-2} + 4$, with $a_0 = 0, a_1 = 1$.

Auxiliary equation and CF. $m^2 - 3m + 2 = (m-1)(m-2) = 0$, so CF = $A + B2^n$.

PI. The extra term is constant, and 1 is a simple auxiliary root. Try Kn . Substitute into the rearranged recurrence:

$$Kn - 3K(n-1) + 2K(n-2) = 4.$$

The left side is $-K$, giving $K = -4$.

Combine. $a_n = A + B2^n - 4n$.

Initial values. At $n = 0$, $A + B = 0$. At $n = 1$, $A + 2B - 4 = 1$, so $A + 2B = 5$. Subtract to get $B = 5$, then $A = -5$.

Answer. $a_n = 5 \cdot 2^n - 4n - 5$.

Practice. What should change if a question says to solve using generating functions?

Solution. Use the requested method: define $A(x)$, multiply by x^n , sum, and solve for $A(x)$. The auxiliary-equation answer can serve as a check, but it does not replace working by the method explicitly requested.

Common error. For a proof question, include the requested justification. For an ordinary solving question, keep the main working visible instead of surrounding it with an unnecessary general proof.

Practice bank: CF and PI practice

Attempt these questions before reading the answers on the following page. For a calculation or proof, write the method and the essential working as well as the final result.

1. Solve $a_n = 2a_{n-1} + 4$, with $a_0 = 3$.
2. Solve $b_n = 3b_{n-1} + 2$, with $b_0 = 0$.
3. Solve $c_n = c_{n-1} + 5$, with $c_0 = 2$.
4. Solve $d_n = 2d_{n-1} + 2^n$, with $d_0 = 0$.
5. Choose a PI trial for an extra term $n^2 + 1$ when 1 is not an auxiliary root.
6. Choose a PI trial for extra term 3^n when 3 is a double auxiliary root.
7. Find a PI for $e_n = 2e_{n-1} + 5^n$.
8. Solve $f_n = 2f_{n-1} + n$, with $f_0 = 0$.

Answers: CF and PI practice

Compare the reasoning as well as the final answer. If your answer differs, return to the first step where the methods separate.

1. $CF = A2^n$. A constant PI satisfies $K = 2K + 4$, so $K = -4$. Thus $a_n = A2^n - 4$. At zero, $A = 7$, giving $7 \cdot 2^n - 4$.
2. The auxiliary root is 3. Try constant PI: $K = 3K + 2$, giving $K = -1$. Then $b_n = A3^n - 1$ and $A = 1$, so $b_n = 3^n - 1$.
3. The CF is a constant. Use $PI = Kn$ and obtain $K = 5$. Hence $c_n = 2 + 5n$.
4. The base 2 matches the auxiliary root, so use $PI = Kn2^n$. Substitution gives $K = 1$. The initial value makes the CF coefficient zero, so $d_n = n2^n$.
5. Use $Kn^2 + Ln + M$. Include all lower powers, even though the forcing has no linear term.
6. Use Kn^23^n . The ordinary trial $K3^n$ needs a factor n^2 for the double match.
7. Try $K5^n$. After substitution and division by 5^{n-1} , $5K = 2K + 5$, giving $K = 5/3$. The PI is $(5/3)5^n$.
8. $CF = A2^n$. The trial $Kn + L$ gives $K = -1, L = -2$. Thus $f_n = A2^n - n - 2$, and the initial value gives $A = 2$. The answer is $f_n = 2^{n+1} - n - 2$.

31 Fibonacci numbers: recognise the recurrence

The Fibonacci numbers begin

$$0, 1, 1, 2, 3, 5, 8, 13, \dots$$

Each new term is the sum of the previous two. With the starting index at zero, write

$$F_0 = 0, \quad F_1 = 1, \quad F_n = F_{n-1} + F_{n-2} \quad (n \geq 2).$$

If the question asks for terms, add the previous two. For example, $F_6 = F_5 + F_4 = 5 + 3 = 8$.

If it asks for a closed formula, use the auxiliary equation:

$$m^2 - m - 1 = 0.$$

The quadratic formula gives roots

$$\phi = \frac{1 + \sqrt{5}}{2}, \quad \psi = \frac{1 - \sqrt{5}}{2}.$$

Write $F_n = A\phi^n + B\psi^n$. The initial values give $A + B = 0$ and $A\phi + B\psi = 1$. Therefore $A = 1/\sqrt{5}$, $B = -1/\sqrt{5}$, and

$$F_n = \frac{\phi^n - \psi^n}{\sqrt{5}}.$$

The Greek letters are only short names for the two roots.

Practice. A staircase can be climbed using steps of size one or two. Find the number of ways to climb four steps.

Solution. Let s_n count the ways. A final one-step move leaves s_{n-1} possibilities; a final two-step move leaves s_{n-2} . Thus $s_n = s_{n-1} + s_{n-2}$, with $s_0 = 1, s_1 = 1$. The counts are 1, 1, 2, 3, 5, so the answer is five.

Common error. Fibonacci-type counts may start with 1, 1 or 1, 2 rather than 0, 1. Use the stated initial values instead of assuming every such count is exactly F_n .

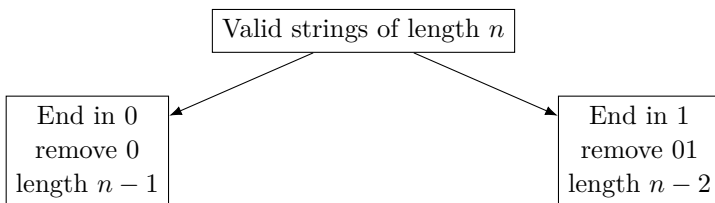
32 Application: binary strings with restrictions

Let b_n count length- n binary strings with no adjacent 1s. The empty string is allowed, so $b_0 = 1$, and the length-one strings are 0, 1, giving $b_1 = 2$.

For $n \geq 2$, a valid string either ends in 0 or in 1. If it ends in 0, removing that last symbol leaves any valid length- $(n - 1)$ string. If it ends in 1, the preceding symbol must be 0; removing the final 01 leaves any valid length- $(n - 2)$ string. Thus

$$b_n = b_{n-1} + b_{n-2}, \quad b_n = F_{n+2}.$$

The recurrence arises from the restriction, not from a numerical coincidence in the first few counts.



The two final-pattern cases give $b_n = b_{n-1} + b_{n-2}$.

Worked example. For length three, the valid strings are 000, 001, 010, 100, 101. Thus $b_3 = 5$, agreeing with $F_5 = 5$. The prohibited strings 011, 110, 111 each contain adjacent 1s.

Practice. Find the number of valid strings of length five. Then explain how the recurrence changes if all binary strings are permitted.

Solution. The restricted values are 1, 2, 3, 5, 8, 13, so $b_5 = 13$. Without the restriction, either final symbol can follow any earlier string. If u_n counts all strings, $u_n = 2u_{n-1}$ with $u_0 = 1$, giving $u_n = 2^n$. The restriction changes the size of the earlier object left after removing a final 1.

Common error. When removing a suffix, prove the reverse construction also works. Here appending 01 to any valid shorter string introduces no adjacent 1s because the appended 0 separates the old end from the new 1.

More complicated forbidden patterns may require several state counts, such as counts indexed by the current suffix. A single total may not contain enough information to describe the next extension correctly.

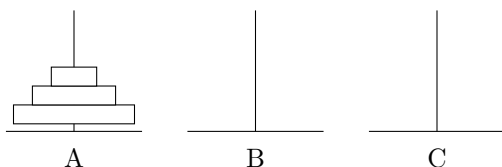
33 Application: the Tower of Hanoi

In the three-peg Tower of Hanoi, n disks of different sizes begin stacked on one peg. A move transfers one top disk, and a larger disk may never rest on a smaller one. Let H_n be the minimum moves needed to transfer the stack to another peg.

To move the largest disk, all $n - 1$ smaller disks must first be on the third peg. This requires at least H_{n-1} moves. Moving the largest disk takes one move. Transferring the smaller stack onto it takes at least another H_{n-1} . The standard recursive procedure achieves exactly these stages, so

$$H_n = 2H_{n-1} + 1, \quad H_0 = 0.$$

Repeated substitution or a constant particular solution gives $H_n = 2^n - 1$.



Practice. Find H_4 and explain why the recurrence establishes a minimum rather than only the cost of one algorithm.

Solution. $H_4 = 2^4 - 1 = 15$. The argument contains a lower bound: any legal solution must clear the largest disk, move it, and then place the smaller disks above it. It also contains a construction achieving that bound. A cost calculation without the necessity argument would establish only an upper bound on the minimum.

Common error. The recurrence relies on exactly three pegs. Adding another peg changes the available strategies, so the same minimum-move proof no longer applies unchanged.

The exponential formula explains the rapid growth: adding one disk doubles the previous task and adds one unavoidable move.

34 Application: regions formed by lines

Let R_n be the largest number of regions into which n lines divide the plane. To achieve the maximum, assume no two lines are parallel and no three pass through the same point.

When the n th line is added, it intersects the previous $n - 1$ lines at distinct points. These points divide the new line into n segments, counting the two unbounded end pieces. Each segment passes through and splits one existing region, so the new line adds n regions. Hence

$$R_n = R_{n-1} + n, \quad R_0 = 1.$$

Summing the increments gives

$$R_n = 1 + \sum_{j=1}^n j = 1 + \frac{n(n+1)}{2}.$$

Worked example. For three lines in general position, $R_3 = 7$. If the three lines meet at one common point, there are only six regions, so the general-position assumptions are doing real work.

Practice. What is the maximum for five lines? What is the region count if all five lines are parallel?

Solution. The maximum is $R_5 = 1 + 5 \cdot 6/2 = 16$. Five distinct parallel lines create six regions, because each added line contributes only one new region. The recurrence for that restricted arrangement is $P_n = P_{n-1} + 1$, with $P_0 = 1$.

Common error. A maximum-count recurrence needs both a limit on how many new regions can be created and an arrangement achieving that limit. The new line has at most $n - 1$ distinct intersections, so it creates at most n new regions; general position achieves this number.

This is an additive recurrence even though the drawing becomes visually complicated. Counting the *change* produced by one new object is often easier than counting the entire resulting arrangement directly.

35 Application: retaining an amount and adding more

A recurrence can describe an amount that changes in stages. Define a_n as the amount after step n . If each step retains a fraction r of the previous amount and then adds c units, write

$$a_n = ra_{n-1} + c.$$

The word then matters: first calculate what remains, and only afterward add the new input.

Worked example. A container retains half its contents, then receives three units. It starts with ten units. Find a formula for the amount after n steps.

- 1. Model the instruction.** The recurrence is $a_n = \frac{1}{2}a_{n-1} + 3$, with $a_0 = 10$.
- 2. Find the CF.** The auxiliary equation is $m - \frac{1}{2} = 0$. Therefore $CF = A(1/2)^n$.
- 3. Find the PI.** Try a constant K . Substitution gives $K = \frac{1}{2}K + 3$, so $K = 6$.
- 4. Use the starting amount.** Now $a_n = A(1/2)^n + 6$. At zero, $10 = A + 6$, so

$$a_n = 4(1/2)^n + 6.$$

The first new amount is eight, agreeing with half of ten plus three. Later amounts get closer to six because the term $4(1/2)^n$ gets smaller.

Practice. If three units are added before half the contents are lost, write the new recurrence.

Solution. The amount before the loss is $b_{n-1} + 3$. Half remains, so $b_n = \frac{1}{2}(b_{n-1} + 3)$. Expanding gives $b_n = \frac{1}{2}b_{n-1} + \frac{3}{2}$.

Practice. With $b_0 = 10$, solve this new recurrence.

Solution. The CF is $A(1/2)^n$. A constant PI satisfies $K = \frac{1}{2}K + \frac{3}{2}$, giving $K = 3$. The initial value gives $A = 7$, so $b_n = 7(1/2)^n + 3$.

Common error. Translating the order of the operations correctly is as important as solving the equation.

Practice bank: Recurrence applications

Attempt these questions before reading the answers on the following page. For a calculation or proof, write the method and the essential working as well as the final result.

1. Count tilings of a length-four strip using pieces of lengths one and two.
2. How many ways can five stairs be climbed using steps of size one or two?
3. Count length-four binary strings with no adjacent ones.
4. Find the minimum Tower of Hanoi moves for five disks and three pegs.
5. Find the maximum number of regions formed by four lines in general position.
6. A quantity begins at ten and increases by four each step. Give its recurrence and formula.
7. An amount is doubled, then increased by one, starting at zero. Find the amount after three steps.
8. Why is the empty tiling counted as one way rather than zero?

Answers: Recurrence applications

Compare the reasoning as well as the final answer. If your answer differs, return to the first step where the methods separate.

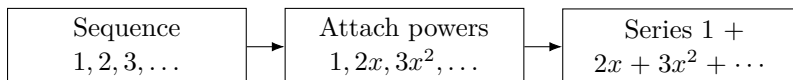
1. Use $t_0 = t_1 = 1$ and $t_n = t_{n-1} + t_{n-2}$. The values are 1, 1, 2, 3, 5, so there are five.
2. The same recurrence gives counts 1, 1, 2, 3, 5, 8, so the answer is eight.
3. Use $b_0 = 1, b_1 = 2$ and $b_n = b_{n-1} + b_{n-2}$. This gives $b_2 = 3, b_3 = 5, b_4 = 8$.
4. Use $H_n = 2^n - 1$. Thus $H_5 = 31$.
5. Use $R_n = 1 + n(n+1)/2$. At four, the answer is $1 + 10 = 11$.
6. $a_0 = 10$ and $a_n = a_{n-1} + 4$. Therefore $a_n = 10 + 4n$.
7. The recurrence is $a_n = 2a_{n-1} + 1$. The values are 0, 1, 3, 7, so the answer is seven.
8. There is one way to use no pieces on a strip of length zero. This supplies the starting count needed when a larger tiling consists of just one final piece.

36 Generating functions: put a sequence into one expression

A generating function writes the sequence as coefficients of powers of x :

$$A(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots$$

The number next to x^n is the term a_n . For these questions, x is a placeholder used for organising the terms. You normally do not substitute a number for x .



An exponent records an index. A coefficient records a value.

Worked example. The sequence 2, 5, 8, 11, ... becomes

$$A(x) = 2 + 5x + 8x^2 + 11x^3 + \dots$$

The first term has no visible x because it multiplies $x^0 = 1$.

Index	0	1	2	3
Sequence term	2	5	8	11
Series term	2	$5x$	$8x^2$	$11x^3$

You may see the shorthand $[x^n]A(x)$. It means the coefficient of x^n in $A(x)$. In the example, $[x^2]A(x) = 8$.

Why use it? Once a recurrence is put into this form, multiplying by x moves its terms one position. This lets us turn the recurrence into an ordinary algebraic equation involving $A(x)$.

Practice. Write the generating function for the sequence 1, 0, 3, 0, 5, 0, ... through the x^5 term.

Solution. It is $1 + 0x + 3x^2 + 0x^3 + 5x^4 + 0x^5 + \dots$, usually shortened to $1 + 3x^2 + 5x^4 + \dots$. Missing powers have coefficient zero.

Practice. What is $[x^3](4 + 2x + 7x^3)$?

Solution. It is 7. You read the coefficient; you do not substitute $x = 3$.

Common error. Keep the starting index at zero unless the question states another convention. Moving the first coefficient to x^1 changes the sequence being represented.

37 The generating-function formulas to learn

Begin with a few familiar sequences. In each row below, the first listed value is at index zero.

Sequence	Term at index n	Generating function
$1, 1, 1, 1, \dots$	1	$1/(1 - x)$
$1, r, r^2, r^3, \dots$	r^n	$1/(1 - rx)$
$1, 2, 3, 4, \dots$	$n + 1$	$1/(1 - x)^2$
$0, 1, 2, 3, \dots$	n	$x/(1 - x)^2$

How to use the table. Split the given sequence formula into familiar pieces. Multiply each generating function by the same constant as its sequence, then add the pieces.

Worked example. Find the generating function for $a_n = 2n + 1$. The $2n$ part gives $2x/(1 - x)^2$. The constant 1 part gives $1/(1 - x)$. Therefore

$$A(x) = \frac{2x}{(1 - x)^2} + \frac{1}{1 - x} = \frac{1 + x}{(1 - x)^2}.$$

Its early coefficients are 1, 3, 5, 7, matching the sequence.

Why the $n + 1$ formula is reasonable. Multiply two copies of $1 + x + x^2 + \dots$. To get x^n , the exponents can be $(0, n), (1, n - 1), \dots, (n, 0)$. There are $n + 1$ choices.

Practice. Find the generating function for $a_n = 4 \cdot 3^n + 2$.

Solution. Use $4/(1 - 3x)$ for the first part and $2/(1 - x)$ for the second. Thus $A(x) = 4/(1 - 3x) + 2/(1 - x)$.

One additional formula. For a positive integer m ,

$$[x^n] \frac{1}{(1 - x)^m} = \binom{n + m - 1}{m - 1}.$$

Here the brackets ask for a coefficient, and the binomial symbol is the combination formula from the prerequisite section. For $m = 3$, it becomes $(n + 1)(n + 2)/2$. The later examples show when to use it.

Common error. The series for n begins with zero. The series for $n + 1$ begins with one. Checking the first coefficient catches this common index error.

38 Shifting a generating function

Multiplication by x moves every coefficient one place:

$$\begin{aligned} A(x) &= a_0 + a_1x + a_2x^2 + \cdots, \\ xA(x) &= a_0x + a_1x^2 + a_2x^3 + \cdots, \\ x^2A(x) &= a_0x^2 + a_1x^3 + a_2x^4 + \cdots. \end{aligned}$$

These lines are the main tool for solving recurrences with generating functions.

For a recurrence starting at $n = 1$:

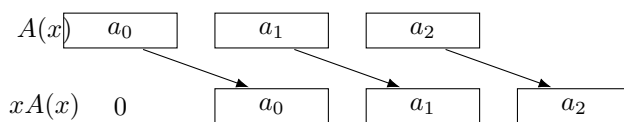
$$\sum_{n \geq 1} a_n x^n = A - a_0, \quad \sum_{n \geq 1} a_{n-1} x^n = xA.$$

The symbol $\sum$ means add all the terms with the indicated indices. Here A is short for $A(x)$.

For a recurrence starting at $n = 2$:

$$\begin{aligned} \sum_{n \geq 2} a_n x^n &= A - a_0 - a_1x, \\ \sum_{n \geq 2} a_{n-1} x^n &= x(A - a_0), \\ \sum_{n \geq 2} a_{n-2} x^n &= x^2A. \end{aligned}$$

Subtract the initial terms that are absent from the required sum.



Multiplying by x shifts all coefficients one place right.

Worked example. If $a_0 = 2$, $a_1 = 5$, then the first of the second-order expressions becomes $A - 2 - 5x$. The second becomes $x(A - 2)$, and the third is still x^2A .

Practice. Write $\sum_{n \geq 2} a_{n-1} x^n$ by expanding its first three terms.

Solution. It is $a_1x^2 + a_2x^3 + a_3x^4 + \cdots$. Factor out x to get $x(a_1x + a_2x^2 + \cdots) = x(A - a_0)$.

Common error. Do not memorise a shift without its starting index. If unsure, write the first two terms and compare them with $A(x)$.

39 Multiplying series to find a coefficient

To find the coefficient of x^n in a product, list the term pairs whose exponents add to n . Multiply their coefficients and add the results.

Worked example. Find the coefficient of x^2 in

$$(1 + 2x + 3x^2)(1 + 4x).$$

The three possible exponent pairs give:

$$1 \cdot (0x^2), \quad (2x)(4x), \quad (3x^2)(1).$$

The first contributes zero because the second bracket has no x^2 term. The other contributions are $8x^2$ and $3x^2$. The required coefficient is $8 + 3 = 11$.

First exponent	Second exponent	Contribution
0	2	0
1	1	$2 \cdot 4 = 8$
2	0	$3 \cdot 1 = 3$

In general, if $A(x)$ has coefficients a_j and $B(x)$ has coefficients b_j , then

$$[x^n]A(x)B(x) = a_0b_n + a_1b_{n-1} + \cdots + a_nb_0.$$

This sum is called convolution. You can solve elementary questions by listing the pairs, without using the name.

Practice. Find $[x^3](1 + x + x^2 + x^3)^2$.

Solution. Use exponent pairs $(0, 3), (1, 2), (2, 1), (3, 0)$. Each contributes one, so the answer is four.

Practice. Find $[x^2](1 + 3x)(1 + 2x + x^2)$.

Solution. The contributions are $1 \cdot 1$ from 1 times x^2 , and $3 \cdot 2$ from $3x$ times $2x$. The answer is $1 + 6 = 7$.

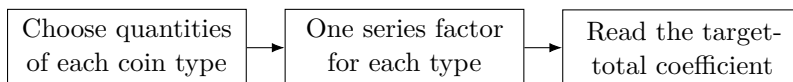
Common error. Do not multiply only the two coefficients at index n . All pairs whose exponents add to n must be considered.

40 Counting with coin-type generating functions

Suppose an item contributes size d and may be chosen any nonnegative number of times. Its possible contributions are represented by

$$1 + x^d + x^{2d} + \cdots = \frac{1}{1 - x^d}.$$

The exponent records total size, while the selected term records the multiplicity of that item. Multiplying one factor for each labelled item type counts combinations of multiplicities.



The product counts combinations of quantities.

Worked example. With denominations 1, 2, 5 and unlimited supplies, the number of combinations totalling n is

$$[x^n] \frac{1}{(1 - x)(1 - x^2)(1 - x^5)}.$$

For $n = 5$, either one 5 is used, giving one combination, or no 5 is used. In the latter case, the number of 2s can be 0, 1, 2, and the remaining amount is filled with 1s. Thus the coefficient is four.

Practice. Count the combinations totalling 6 using unlimited 1, 2, and 3 units.

Solution. The generating function is $1/((1 - x)(1 - x^2)(1 - x^3))$. Condition on the number c of 3-unit items. If $c = 0$, the number of 2s can be 0, 1, 2, 3, giving four choices. If $c = 1$, the remaining total is 3, giving two choices for the number of 2s. If $c = 2$, only one combination remains. The total is $4 + 2 + 1 = 7$.

Common error. This product counts combinations, not ordered sequences of coin placements. A 1 followed by a 2 and a 2 followed by a 1 have the same multiplicities and therefore contribute one combination, not two.

If order matters, a different construction is needed. For sequences of parts of sizes 1 and 2, the generating function is $1/(1 - x - x^2)$, because each new part may be chosen in either size. The distinction between a product over types and a sequence of choices is essential.

41 Bounded selections and minimum requirements

Generating functions encode restrictions directly in their factors. If a type of size d may occur at most b times, use $1 + x^d + \cdots + x^{bd}$. If it must occur at least a times with no upper bound, use $x^{ad}/(1 - x^d)$.

Worked example. Count integer solutions of $u + v + w = 6$ with $0 \leq u \leq 2$, $v \geq 1$, and $w \geq 0$. The generating function is

$$(1 + x + x^2) \frac{x}{1 - x} \frac{1}{1 - x}.$$

The desired coefficient is $[x^5](1 + x + x^2)/(1 - x)^2$. Since $[x^m](1 - x)^{-2} = m + 1$, it equals $6 + 5 + 4 = 15$. These three contributions correspond to $u = 0, 1, 2$.

Practice. How many solutions satisfy $u + v + w = 5$ with $0 \leq u, v \leq 2$ and $w \geq 0$?

Solution. Use $(1 + x + x^2)^2/(1 - x)$. Every one of the nine ordered pairs (u, v) has $u + v \leq 4$, so exactly one nonnegative $w = 5 - u - v$ completes it. Hence the coefficient is nine. The generating function records the same reasoning: summing the coefficients of the bounded product through degree five includes all its nine choices.

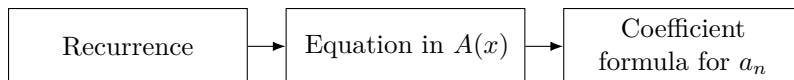
Common error. A lower bound is implemented by a shift, not by deleting an arbitrary number of coefficients from the final product. For $v \geq 1$, the factor begins with x , whereas an unrestricted variable begins with 1.

For a finite geometric factor, the identity $1 + x + \cdots + x^b = (1 - x^{b+1})/(1 - x)$ can make coefficient extraction easier. The numerator subtracts precisely those choices that exceed the bound, connecting generating functions with inclusion–exclusion.

Always define whether the variables are labelled. In these equations, (u, v, w) is ordered because the three positions have distinct roles.

42 Generating-function method: a first-order question

Use this method when the question specifically asks for generating functions. The answer steps are: define $A(x)$, multiply by x^n , add the equations, solve for $A(x)$, and read its coefficients.



Multiply, sum, rearrange, and extract.

Worked example. Solve $a_n = 2a_{n-1} + 1$ for $n \geq 1$, with $a_0 = 0$.

1. Define the series. Let $A(x) = a_0 + a_1x + a_2x^2 + \cdots$.

2. Multiply and add. Multiplying the recurrence by x^n and adding for $n \geq 1$ gives

$$A - a_0 = 2xA + (x + x^2 + x^3 + \cdots).$$

Use $a_0 = 0$ and $x + x^2 + \cdots = x/(1 - x)$:

$$A = 2xA + \frac{x}{1 - x}.$$

3. Solve the algebra.

$$A(1 - 2x) = \frac{x}{1 - x}, \quad A = \frac{x}{(1 - x)(1 - 2x)}.$$

4. Split into familiar forms.

$$A = \frac{1}{1 - 2x} - \frac{1}{1 - x}.$$

The coefficients of these two series are 2^n and 1. Therefore $a_n = 2^n - 1$.

Practice. Solve $b_n = 3b_{n-1}$ with $b_0 = 2$ by this method.

Solution. Let its series be $B(x)$. Summing from $n = 1$ gives $B - 2 = 3xB$. Thus $B = 2/(1 - 3x)$, whose coefficient of x^n is $2 \cdot 3^n$. Hence $b_n = 2 \cdot 3^n$.

Common error. The first sum starts at $n = 1$, so the left side is $A - a_0$. Including or excluding the wrong initial term changes the answer.

43 Generating-function method: a second-order question

For a second-order recurrence, begin the summation at $n = 2$ when a_0, a_1 are the given starting values. Keep the shift table beside your working.

Worked example. Solve $a_n = 3a_{n-1} - 2a_{n-2}$, with $a_0 = 0, a_1 = 1$.

1. Write the summed equation.

$$A - a_0 - a_1x = 3x(A - a_0) - 2x^2A.$$

The three terms come directly from the second-order shift table.

2. Put in the initial values.

$$A - x = 3xA - 2x^2A.$$

3. Collect the terms containing A .

$$A(1 - 3x + 2x^2) = x.$$

Factor the bracket:

$$A = \frac{x}{(1-x)(1-2x)} = \frac{1}{1-2x} - \frac{1}{1-x}.$$

4. Read the coefficient. The answer is $a_n = 2^n - 1$.

Practice. Find the generating function for $b_n = b_{n-1} + b_{n-2}$ with $b_0 = 0, b_1 = 1$.

Solution. The summed equation is $B - x = xB + x^2B$. Therefore $B(1 - x - x^2) = x$, giving $B = x/(1 - x - x^2)$. This is the generating function for the Fibonacci sequence.

Optional shortcut after practising the method. For $a_n = c_1a_{n-1} + c_2a_{n-2}$,

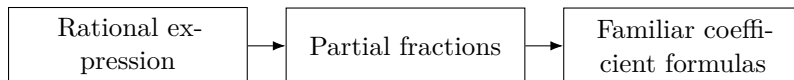
$$A(x) = \frac{a_0 + (a_1 - c_1a_0)x}{1 - c_1x - c_2x^2}.$$

The letters c_1, c_2 are the multipliers in the recurrence.

Common error. The numerator depends on the initial values. The recurrence alone gives the denominator, not the complete generating function.

44 Partial fractions: how to split the answer

Partial fractions turn a generating function into pieces whose coefficients you know. If the denominator is $(1-x)(1-2x)$, try one unknown constant over each bracket.



Split the fraction before reading the sequence.

Worked example. Split

$$\frac{x}{(1-x)(1-2x)} = \frac{C}{1-x} + \frac{D}{1-2x}.$$

Multiply both sides by the common denominator:

$$x = C(1-2x) + D(1-x).$$

Put $x = 1$: $1 = -C$, so $C = -1$. Put $x = 1/2$: $1/2 = D/2$, so $D = 1$. Therefore

$$\frac{x}{(1-x)(1-2x)} = -\frac{1}{1-x} + \frac{1}{1-2x}.$$

The coefficient of x^n is $-1 + 2^n$.

Here substituting $x = 1$ or $1/2$ is done in the polynomial equation *after clearing denominators*. Do not substitute these values directly into a fraction whose denominator becomes zero.

Practice. Find $[x^n]1/((1-x)(1-2x))$.

Solution. Write $1 = C(1-2x) + D(1-x)$. At $x = 1$, $C = -1$. At $x = 1/2$, $D = 2$. Thus

$$\frac{1}{(1-x)(1-2x)} = -\frac{1}{1-x} + \frac{2}{1-2x},$$

so the coefficient is $2^{n+1} - 1$.

Common error. Keep the numerator when clearing denominators. A numerator x and a numerator 1 lead to different constants, even though the denominator is the same.

Practice bank: Generating functions and partial fractions

Attempt these questions before reading the answers on the following page. For a calculation or proof, write the method and the essential working as well as the final result.

1. Write the generating function for $1, 1, 1, \dots$
2. Write the generating function for $1, 4, 16, 64, \dots$
3. Find $[x^6](1-x)^{-2}$.
4. Find $[x^6]x^2/(1-x)^2$.
5. Find $[x^3](1+2x)(1+x+x^2+x^3)$.
6. Split $1/((1-x)(1-3x))$ into partial fractions.
7. Find the coefficient at index n in the previous fraction.
8. For a second-order recurrence summed from $n = 2$, express $\sum_{n \geq 2} a_n x^n$.

Answers: Generating functions and partial fractions

Compare the reasoning as well as the final answer. If your answer differs, return to the first step where the methods separate.

1. $1/(1-x)$, since its expansion is $1 + x + x^2 + \dots$.
2. $1/(1-4x)$. Its coefficient at index n is 4^n .
3. The coefficient is $n+1$ at index n , so the answer is seven.
4. Subtract the numerator shift: use index four in $(1-x)^{-2}$. The answer is five.
5. The contributions are 1 times x^3 and $2x$ times x^2 . Their coefficients add to three.
6. Write $1 = C(1-3x) + D(1-x)$. Substituting $x = 1$ gives $C = -1/2$; $x = 1/3$ gives $D = 3/2$. The split is $-\frac{1}{2(1-x)} + \frac{3}{2(1-3x)}$.
7. Read the coefficients of the split terms: $-\frac{1}{2} + \frac{3}{2} 3^n = (3^{n+1} - 1)/2$.
8. It is $A(x) - a_0 - a_1x$. Remove the two starting terms excluded from the sum.

45 Repeated denominator factors: the useful formulas

For repeated factors, use these coefficient formulas:

$$\begin{aligned}[x^n] \frac{1}{1-rx} &= r^n, \\ [x^n] \frac{1}{(1-rx)^2} &= (n+1)r^n, \\ [x^n] \frac{1}{(1-x)^3} &= \frac{(n+1)(n+2)}{2}.\end{aligned}$$

The notation $[x^n]$ means take the coefficient of x^n . The constant r is the multiplier of x in the denominator.

Worked example. Find $[x^4](1-2x)^{-2}$. Use the second formula with $n=4, r=2$: the answer is $(4+1)2^4 = 80$.

If there is an x in the numerator, shift the index first. For example,

$$[x^n] \frac{x}{(1-x)^2} = n \quad (n \geq 1).$$

The coefficient needed from $(1-x)^{-2}$ is at index $n-1$, giving $(n-1)+1 = n$.

Practice. Find $[x^5]x^2/(1-x)^3$.

Solution. The x^2 uses two powers, so take the coefficient of x^3 in $(1-x)^{-3}$. It is $(3+1)(3+2)/2 = 10$.

Practice. Find $[x^n](1+x)/(1-x)^2$.

Solution. The 1 part contributes $n+1$, and the x part contributes n for $n \geq 1$. The answer is $2n+1$, also correct at $n=0$ because the constant coefficient is 1.

Common error. When splitting a repeated factor into partial fractions, include every required power. For a factor $(1-rx)^2$, use both $C/(1-rx)$ and $D/(1-rx)^2$ if needed.

46 Generating functions with an extra n term

An extra term in a recurrence contributes its own generating function. Use the standard formula

$$nx^n \text{ summed for } n \geq 1 \longrightarrow \frac{x}{(1-x)^2}.$$

The method otherwise stays the same.

Worked example. Solve $a_n = 2a_{n-1} + n$, with $a_0 = 0$.

1. Multiply by x^n and sum for $n \geq 1$.

$$A = 2xA + \frac{x}{(1-x)^2}.$$

2. Solve for A .

$$A = \frac{x}{(1-2x)(1-x)^2}.$$

3. Use partial fractions.

$$A = \frac{2}{1-2x} - \frac{1}{1-x} - \frac{1}{(1-x)^2}.$$

You can check the split by multiplying both sides by the common denominator.

4. Read each coefficient. The three pieces give $2 \cdot 2^n$, -1 , and $-(n+1)$. Hence

$$a_n = 2^{n+1} - n - 2.$$

At $n = 0$, the formula gives zero, as required.

Practice. Solve $b_n = b_{n-1} + 1$ with $b_0 = 0$ using generating functions.

Solution. Let its series be B . Summation gives $B = xB + x/(1-x)$. Thus $B = x/(1-x)^2$, whose coefficient is $b_n = n$.

Common error. If the recurrence starts at $n = 2$, the extra-term series must also start there. For example, $\sum_{n \geq 2} nx^n = x/(1-x)^2 - x$, because the $n = 1$ term must be removed.

47 Generating functions when the order of pieces matters

Suppose a construction is an ordered list of pieces. First write one expression for the allowed choices of a single piece. Call it $C(x)$. Then zero pieces contribute 1, one piece contributes $C(x)$, two pieces contribute $C(x)^2$, and so on.

$$1 + C(x) + C(x)^2 + \cdots = \frac{1}{1 - C(x)}.$$

This is the geometric-series formula with $C(x)$ in place of x . We use pieces of positive size, so a fixed total cannot contain endlessly many pieces.

Worked example. A strip is tiled with pieces of lengths one and two. One piece contributes $C(x) = x + x^2$. Hence

$$T(x) = \frac{1}{1 - x - x^2}.$$

For a total length of three, the possibilities are $(1, 1, 1)$, $(1, 2)$, $(2, 1)$. The last two are different because their order differs.

Choosing the correct construction.

- If only quantities of each type matter, multiply one factor for each type.
- If the order of successive pieces matters, use the sequence expression above.

Practice. Find a generating function for ordered sums using only parts 1 and 3. Find the coefficient of x^4 .

Solution. One part contributes $x + x^3$, so the generating function is $1/(1 - x - x^3)$. Total four can be made as $(1, 1, 1, 1)$, $(1, 3)$, $(3, 1)$. Thus the coefficient is three.

Practice. If order is ignored in that question, how many choices remain?

Solution. Two: four parts of size one, or one part of each size. The generating function is $1/((1 - x)(1 - x^3))$, since it records only the quantities of the two sizes.

Common error. Decide whether $(1, 3)$ and $(3, 1)$ count as different before choosing the formula. The algebra must reflect the outcomes described in the question.

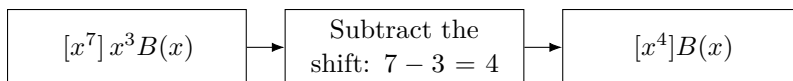
48 Coefficient shortcuts for exam questions

Many generating-function questions become short once you recognise a standard denominator and account for any power of x in the numerator.

Keep these three patterns available.

$$\begin{aligned}\frac{1}{1 - rx} &\longrightarrow r^n, \\ \frac{1}{(1 - x)^2} &\longrightarrow n + 1, \\ \frac{1}{(1 - x)^3} &\longrightarrow \frac{(n + 1)(n + 2)}{2}.\end{aligned}$$

The expression on the right is the coefficient at index n .



The numerator uses three of the required powers.

Worked example. Find $[x^7]x^3/(1 - x)^2$. The numerator already contributes x^3 , so the denominator series must contribute x^4 . Its coefficient is $4 + 1 = 5$.

Adding a sequence cumulatively. If $A(x)$ represents $a_0, a_1, a_2, \dots$, then $A(x)/(1 - x)$ represents

$$a_0, \quad a_0 + a_1, \quad a_0 + a_1 + a_2, \dots$$

For example, the series for $1, 2, 3, \dots$ is $(1 - x)^{-2}$. Its cumulative sums $1, 3, 6, 10, \dots$ have series $(1 - x)^{-3}$.

Practice. Find $[x^6]3x^2/(1 - 2x)$.

Solution. Take the coefficient at index $6 - 2 = 4$ in $(1 - 2x)^{-1}$, then multiply by 3. The answer is $3 \cdot 2^4 = 48$.

Practice. Find $[x^2]x^3/(1 - x)$.

Solution. Zero. The lowest power in the expression is x^3 , so there is no x^2 term.

Common error. A numerator shift changes the index used in the entire coefficient formula. For $x^2/(1 - x)^3$, use $n - 2$ wherever n occurs in $(n + 1)(n + 2)/2$.

49 Choosing between the two recurrence methods

Use the method named in the question. If no method is specified, an auxiliary equation is often the shortest route for a linear recurrence with constant coefficients. Generating functions are especially useful when the question asks for a series or a coefficient.

Worked example. Solve $a_n = 3a_{n-1} - 2a_{n-2}$, with $a_0 = 0, a_1 = 1$.

Auxiliary-equation route. Factor $m^2 - 3m + 2 = (m - 1)(m - 2)$. Write $a_n = A + B2^n$. The starting values give $A + B = 0$ and $A + 2B = 1$, so $a_n = 2^n - 1$.

Generating-function route. The shift formulas give

$$A(x) - x = 3xA(x) - 2x^2A(x).$$

Hence $A(x) = x/((1-x)(1-2x))$. Split it into $1/(1-2x) - 1/(1-x)$ and read the coefficient $2^n - 1$.

Both methods give the same answer. Practise recognising which route the wording requires.

Practice. Which method would you choose for each question?

- (a) Solve $b_n = 5b_{n-1}$ with $b_0 = 2$.
- (b) Find the coefficient of x^{10} in $(1-x)^{-2}$.
- (c) Solve a recurrence using generating functions.

Solution. (a) Use $m-5=0$, giving $b_n = 2 \cdot 5^n$. (b) Use the coefficient formula, giving 11. (c) Show the generating-function working; an auxiliary-equation calculation alone would not meet the method requirement.

Common error. Do not carry out both methods in an exam unless asked. One complete, correct solution is better than two unfinished ones.

50 Unit 3 exam method sheet

Pigeonhole questions. Identify the objects and categories. Divide the number of objects by the number of categories and round upward. For at least r in one of k categories, the guarantee threshold is $k(r - 1) + 1$.

Homogeneous recurrence questions.

1. Rearrange with zero on the right.
2. Form and solve the auxiliary equation.
3. Different roots: add their exponential terms. Repeated root: include powers of n .
4. Substitute initial values to find the constants.

Nonhomogeneous recurrence questions. Find CF and PI separately. Choose the PI trial from the extra term. Test for a matching auxiliary root and multiply the trial by the necessary power of n . Add CF and PI before using starting values.

Generating-function questions. Define $A(x)$, use the correct shifted sums, solve for $A(x)$, and use familiar series or partial fractions to read the coefficient.

Practice. What is the answer form for a double auxiliary root 3?

Solution. $(A + Bn)3^n$.

Practice. For $a_n = 2a_{n-1} + 2^n$, what PI trial should you use?

Solution. $Kn2^n$, because 2 is a simple root of $m - 2 = 0$.

Practice. What does $[x^5]A(x)$ ask for?

Solution. The coefficient multiplying x^5 , not the value obtained by putting $x = 5$.

Final checks. Keep the signs when moving recurrence terms. Use the actual starting indices. In counting, decide whether order matters. In generating functions, subtract any initial terms excluded by the summation limit.

The cumulative questions that follow are a way to practise choosing and executing the method. No particular mark allocation or university paper pattern is assumed.

Practice bank: Mixed Unit 3 revision

Attempt these questions before reading the answers on the following page. For a calculation or proof, write the method and the essential working as well as the final result.

1. Find $[x^4](1 - 2x)^{-2}$.
2. Find $[x^5]x/(1 - x)^3$.
3. Count nonnegative solutions of $u + v = 7$.
4. Count nonnegative solutions of $u + v + w = 4$.
5. Count nonnegative solutions of $u + v + w = 4$ with $u \leq 1$.
6. Find a_3 for $a_n = 3a_{n-1} - 2a_{n-2}$, $a_0 = 0$, $a_1 = 1$.
7. How many people guarantee five share one of four categories?
8. What PI trial fits $n + 2^n$ if the auxiliary roots are 1, 2, both simple?

Answers: Mixed Unit 3 revision

Compare the reasoning as well as the final answer. If your answer differs, return to the first step where the methods separate.

1. Use $(n+1)2^n$ at $n=4$. The answer is $5 \cdot 16 = 80$.
2. Use index four in $(1-x)^{-3}$. The answer is $(4+1)(4+2)/2 = 15$.
3. Choose u from zero through seven; each choice fixes v . There are eight solutions, also the coefficient of x^7 in $(1-x)^{-2}$.
4. Use the coefficient of x^4 in $(1-x)^{-3}$: $(5)(6)/2 = 15$.
5. Use $(1+x)/(1-x)^2$. The coefficient is $5+4=9$, corresponding to $u=0$ or $u=1$.
6. The terms are $a_2=3$ and $a_3=9-2=7$. The formula 2^n-1 gives the same answer.
7. $4(5-1)+1=17$. Sixteen could be split as four per category.
8. Use $n(Kn+L)+Mn2^n$. Each part has one matching root and therefore needs one extra factor n .

51 Cumulative practice questions

Attempt these questions before reading the next section. Include the model, method, and verification, rather than only the final number or formula.

1. What is the smallest number of objects that guarantees seven in one of nine boxes? Justify minimality.
2. Prove that among any eight integers, two have a difference divisible by seven.
3. Given five integers, prove that some nonempty consecutive block has sum divisible by five.
4. Solve $a_n = 4a_{n-1} - 3a_{n-2}$ with $a_0 = 2, a_1 = 4$.
5. Solve $b_n = 6b_{n-1} - 9b_{n-2}$ with $b_0 = 1, b_1 = 6$.
6. Solve $c_n = 2c_{n-1} + 2^n$ with $c_0 = 0$.
7. Solve $d_n = d_{n-1} + 3n + 1$ with $d_0 = 2$.
8. Derive a recurrence and initial values for length- n binary strings with no adjacent 0s. Find the count at $n = 4$.
9. Find $[x^n](1+x)/(1-x)^2$.
10. How many nonnegative integer solutions satisfy $u + v + w = 8$ and $u \leq 2$? Give a generating-function solution.
11. Find the generating function of $e_n = 3e_{n-1} - 2e_{n-2}$ with $e_0 = 1, e_1 = 4$, then extract e_n .
12. Find the coefficient of x^5 in $1/(1-x-x^2)$ and explain its tiling meaning.

Working guidance. In Questions 1–3, identify the counted objects and their boxes explicitly. In Questions 4–7, find the CF and PI separately and keep the starting index visible. In Question 8, justify how deleting a final symbol or suffix maps to smaller valid strings. In Questions 9–12, state which coefficient identity or shift you use.

Extension. For Question 10, solve the problem again by summing separately over $u = 0, 1, 2$. Agreement between the direct count and generating function provides a useful independent check.

Do not confuse a numerical check with a proof. Testing the first three terms helps catch algebraic mistakes, while satisfying the recurrence symbolically and matching all initial values proves the closed formula.

52 Cumulative solutions and further reading

Solutions 1–6

1. The largest avoiding arrangement has six objects in each of nine boxes, for 54 total. Therefore 55 guarantee seven. The occupancy vector $(6, 6, \dots, 6)$ shows that 54 do not suffice.
2. The eight integers occupy seven remainder classes modulo 7. Two have the same remainder, so their difference is divisible by 7.
3. Form six partial sums $S_0 = 0, S_1, \dots, S_5$. Two have the same remainder modulo 5. If their indices are $i < j$, the nonempty block from a_{i+1} through a_j sums to $S_j - S_i$, a multiple of 5.
4. The auxiliary equation is $(m - 1)(m - 3) = 0$. Write $a_n = A + B3^n$. The equations $A + B = 2$ and $A + 3B = 4$ give $A = B = 1$. Thus $a_n = 1 + 3^n$.
5. The auxiliary equation is $(m - 3)^2 = 0$, so $b_n = (A + Bn)3^n$. At $n = 0$, $A = 1$. At $n = 1$, $3(1 + B) = 6$, giving $B = 1$. Hence $b_n = (n + 1)3^n$.
6. The auxiliary equation is $m - 2 = 0$. The extra term 2^n matches its root, so try $\text{PI} = Kn2^n$. Substitution gives $K = 1$. The CF is $C2^n$. Since $c_0 = 0$, $C = 0$, so $c_n = n2^n$. Direct substitution gives $2(n - 1)2^{n-1} + 2^n = n2^n$.

Solutions 7–12

7. The auxiliary equation is $m - 1 = 0$, giving a constant CF A . The extra term $3n + 1$ suggests a linear trial, but the root 1 requires a factor of n . Use $PI = Kn^2 + Ln$. Its difference from the previous term is

$$K[n^2 - (n - 1)^2] + L[n - (n - 1)] = 2Kn + (L - K).$$

Match this with $3n + 1$: $2K = 3$ and $L - K = 1$, so $K = 3/2, L = 5/2$. Now $d_n = A + (3/2)n^2 + (5/2)n$. The initial value gives $A = 2$, hence $d_n = (3n^2 + 5n + 4)/2$.

8. Let $s_0 = 1, s_1 = 2$. For $n \geq 2$, strings ending in 1 correspond to valid length- $(n - 1)$ strings; strings ending in 0 must end in 10 and correspond to length- $(n - 2)$ strings. Thus $s_n = s_{n-1} + s_{n-2}$, giving $s_2 = 3, s_3 = 5, s_4 = 8$.
9. The unshifted part contributes $n + 1$. The shifted part contributes n for $n \geq 1$ and zero at $n = 0$. Therefore the coefficient is $2n + 1$ for every $n \geq 0$.
10. Use $(1 + x + x^2)/(1 - x)^2$. The coefficient of x^8 is $9 + 8 + 7 = 24$. Directly, fixing $u = 0, 1, 2$ leaves $v + w = 8, 7, 6$, with respectively 9, 8, 7 nonnegative solutions.
11. The second-order formula gives

$$E(x) = \frac{1 + (4 - 3)x}{1 - 3x + 2x^2} = \frac{1 + x}{(1 - x)(1 - 2x)}.$$

Partial fractions give $-2/(1 - x) + 3/(1 - 2x)$, so $e_n = 3 \cdot 2^n - 2$. The initial values are 1 and 4, as required.

12. The coefficients satisfy $t_0 = t_1 = 1$ and $t_n = t_{n-1} + t_{n-2}$. Thus the first six are 1, 1, 2, 3, 5, 8, so the coefficient is eight. It counts ordered tilings of a length-five strip by squares and dominoes, including all possible positions of each piece.

Further reading and scope

These explanations and exercises are original instructional material covering the supplied Unit 3 syllabus. For additional problems and broader context, consult Eric Lehman, F. Thomson Leighton, and Albert R. Meyer, *Mathematics for Computer Science*, through MIT OpenCourseWare. Relevant themes include counting, pigeonhole arguments, recurrences, and ordinary generating functions.

The main recurrence methods here assume linear relations with constant coefficients. Variable-coefficient, nonlinear, and divide-and-conquer recurrences require additional techniques. In the generating-function questions, read coefficients from the series; numerical substitution for x is not required. Keeping these boundaries clear helps identify when a familiar method applies and when a new argument is needed.